

Face numbers
of polytopes

Let $f_i(P)$ be the number of i -dim faces of a d -polytope P .

The boundary of P is a $(d-1)$ -sphere, so Euler's relation holds:

$$f_0 - f_1 + f_2 - \dots + (-1)^{d-1} f_{d-1} = \chi(\partial P)$$

If P is **simplizial**, i.e. every face besides P is a simplex, then the Dehn-Sommerville equations hold.

Dehn-Somerville equations:

If Δ is a simplicial d -manifold, then for all $0 \leq k \leq d-1$

$$\sum_{j=0}^d (-1)^j \binom{j+1}{k+1} f_j = (-1)^d f_k$$

and

$$\sum_{j=0}^d (-1)^j f_j = \chi(\Delta)$$

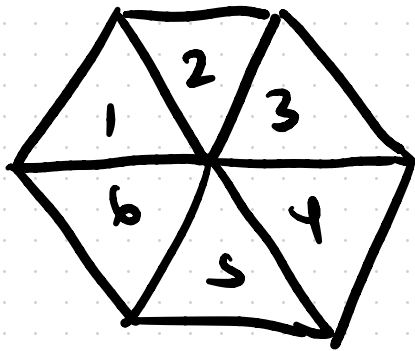
A simplicial complex is **pure** if all of its maximal cells have the same dimension.

A **shelling** of a pure d -dimensional simplicial complex is an ordering $F_1, F_2, \dots, F_n$ of the maximal cells such that for all $1 \leq j \leq n$,

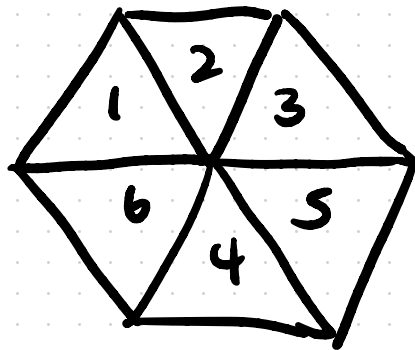
$$F_j \cap \left(\bigcup_{i=1}^{j-1} F_i \right)$$

is empty or a pure complex of dimension $d-1$.

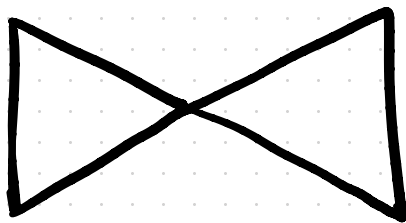
A complex is **shellable** if it has a shelling.



shelling



not a shelling



not shellable

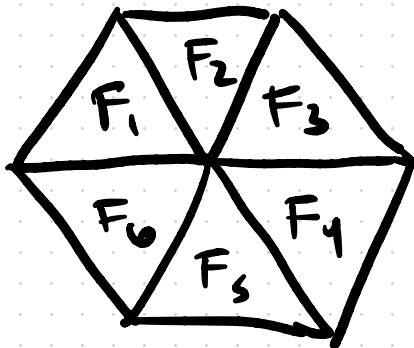
Prop: If Δ is shellable and dimension d , then every connected component of Δ is contractible or homotopy equivalent to a wedge of d -spheres.

Given a shelling $F_1, \dots, F_n$ of Δ , let S_k be the set of all F_j such that

$$F_j \cap \left(\bigcup_{i=1}^{j-1} F_i \right)$$

$$\overset{\uparrow}{0 \leq k \leq d+1}$$

has k maximal faces.



$$S_0 = \{F_1\}$$

$$S_1 = \{F_2, \dots, F_5\}$$

$$S_2 = \{F_6\}$$

$$S_3 = \emptyset$$

Prop: Let $(f_{-1}, f_0, f_1, \dots, f_d)$ be the f -vector of Δ .

Assume Δ is connected. Then

$$f_i = \sum_{k=0}^{d+1} \binom{d+1-k}{i+1-k} |S_k|$$

Proof: Keep track of how many faces of each dimension are added at each step of the shelling.

For any simplicial complex Δ of dimension d , let $h_0(\Delta), h_1(\Delta), \dots, h_{d+1}(\Delta)$ be the integers such that

$$\sum_{i=-1}^d f_i(\Delta) (x-1)^{d-i} = \sum_{i=0}^{d+1} h_i(\Delta) x^{d+1-i}$$

$$\sum_{i=-1}^{d+1} f_i x^{d-i} = \sum_{i=0}^{d+1} h_i (x+1)^{d+1-i}$$

$$f_i = [x^{d-i}] \sum_{k=0}^{d+1} h_k (x+1)^{d+1-k}$$

$$= \sum_{k=0}^{d+1} \binom{d+1-k}{d-i} h_k$$

$$= \sum_{k=0}^{d+1} \binom{d+1-k}{i+1-k} h_k$$

Thm: $|S_k| = h_k$, where $(h_0, \dots, h_{d+1})$ is the h -vector of Δ .

Cor: $|S_k|$ only depends on Δ .

Cor: $h_k(\Delta) \geq 0$ if Δ has a shelling.

Thm: If Δ is the boundary of a simplicial polytope, then it has a shelling.